ARTICLE

PRISM: a modular system for glioma progression surveillance on MRI

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ARTICLE HISTORY

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ABSTRACT

[PROPOSED, awaiting approval] Models of post-treatment glioma MRI face known challenges: most datasets lack treatment dates, centres document progression differently, cohorts are small and the hardest situations have few cases. We built PRISM, a modular system that keeps image and treatment-timeline evidence in separate pathways, measures the tumour by automatic segmentation, attaches a confidence to every scan and writes a report with a local language model. On MU-Glioma-Post, the timeline pathway detected the scan at which progression was called with an AUC of 0.82 and the image pathway with 0.71. Fitted once and locked, the models reached 0.77 on LUMIERE, and single-scan images reached 0.80 on UCSF-ALPTDG, about as high as models trained there. An audit showed what each score rested on: the treatment timeline at MU, enhancing tumour volume at UCSF. Conformal sets kept their coverage at all three centres, and every evaluation report passed a deterministic check against its record.

KEYWORDS

glioma surveillance; external validation; clinical decision support

1. Introduction

After surgery and chemoradiotherapy, patients with glioma are followed with MRI every few months, and at each scan the team must decide whether the tumour has progressed.

Models for this task meet five challenges that come from the data rather than from any one method.

We asked whether a system could give the clinician, at each post-treatment scan, the image evidence and the treatment-timeline evidence separately, carry to other centres without refitting, and say how far each can be trusted there.

PRISM is a modular system built for this, and we tested it on one development cohort and two external cohorts with models fitted once and plans written in advance.

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2. Methods

2.1. PRISM

PRISM handles each scan in five modules, each answering one of the challenges above (Figure 1).

The timeline pathway scores a scan from clinical data and the patient's position in treatment.

The image pathway scores a scan from simple measurements of the segmented tumour.

A deep segmentation network supplies these measurements and the images shown in the report.

Each scan's confidence has three parts: a conformal set, a pathway-agreement statement and a reliability card.

A local language model writes the report from the scan's record, and a deterministic checker tests every sentence against that record.

2.2. Cohorts and the task

PRISM was developed on MU-Glioma-Post and tested, locked, on LUMIERE and UCSF-ALPTDG (Table 1).

The task is detection at the scan: is this the scan at which progression is called?

2.3. Evaluation

On MU, every model was evaluated by repeated patient-grouped cross-validation, because a single split moved scores more than any model choice (Supplementary Figure [S-split]).

For the external tests, each model was fitted once on MU and applied once, under plans committed before any external label was read.

Analyses that were not planned are labelled exploratory wherever they appear.

3. Results

3.1. Each pathway carries evidence on MU-Glioma-Post

In the intended use on MU-Glioma-Post, the timeline pathway detected the scan at which progression was called with an AUC of 0.82, and the image pathway with 0.71.

3.2. The pathways hold at two external centres

Fitted once on MU and applied unchanged, the combined model reached 0.77 on LUMIERE, and single-scan image measurements reached 0.80 on UCSF-ALPTDG (Figure 2).

3.3. What each score rests on depends on the cohort

The audit shows that on MU the scores rest mainly on where the patient is in treatment, and on UCSF on how much the tumour enhances (Figure 3).

Each step of the image model's design raised its AUC on MU, and the audit shows what each step added.

3.4. The confidence keeps its coverage at all three centres

At the 80% level, the conformal sets of the combined score covered 0.82 of MU scans, 0.87 of LUMIERE scans and 0.78 of UCSF scans, at a nominal level of 0.80 and a threshold of 0.78 fixed in advance.

When the two pathways disagreed on MU, the combined score was less accurate, and setting those scans aside raised the AUC of the rest more than removing the same number at random.

For most scans the evidence behind any score is thin, and the reliability card says so.

3.5. The reports are faithful to the record

Every evaluation report written by the local model passed the deterministic check against its record: 114 of 114 on MU, 162 of 162 on LUMIERE and 130 of 130 on UCSF (Figure 5).

4. Discussion

PRISM gives the clinician separate image and treatment-timeline evidence at each scan, carries to two external centres without refitting, and states how far its scores can be trusted there.

Because its modules are separate, PRISM runs where data are incomplete.

Consider a scan eight weeks after radiotherapy in which the enhancing tumour has grown.

Each of PRISM's components has been described before, but we found no system that combines them.

This study has limits.

The next step is to pool the rare situations across cohorts and train on them.

Conclusion

PRISM shows that a surveillance system can keep its kinds of evidence apart, carry them to other centres and say where they have and have not been shown to work.

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Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other open-weight mod-

els in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no patient-level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

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Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of de-identified datasets (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

References

Suter Y, Knecht U, Valenzuela W et al. 2022. The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data. 9(1):768. https://doi.org/10.1038/s41597-022-01881-7

Tables

Table 1. The three cohorts. [SKELETON: per cohort, what it has (images, treatment dates, how progression is documented), patients, scans in the detection task, positives and negatives; grade 4 primary. Sources: TABLE1_COHORT.md (MU, LUMIERE), UCSF_DATA.md.]

Table 2. Questions fixed in advance and their answers. [SKELETON: MU detection (DETECTION_TASK.md), LUMIERE, UCSF Q1–Q4 (UCSF_PLAN.md), system S1–S7 (SYSTEM_PLAN.md, REPORT_V3_1_PLAN.md); each with its rule, result and met / not met.]

Figure legends

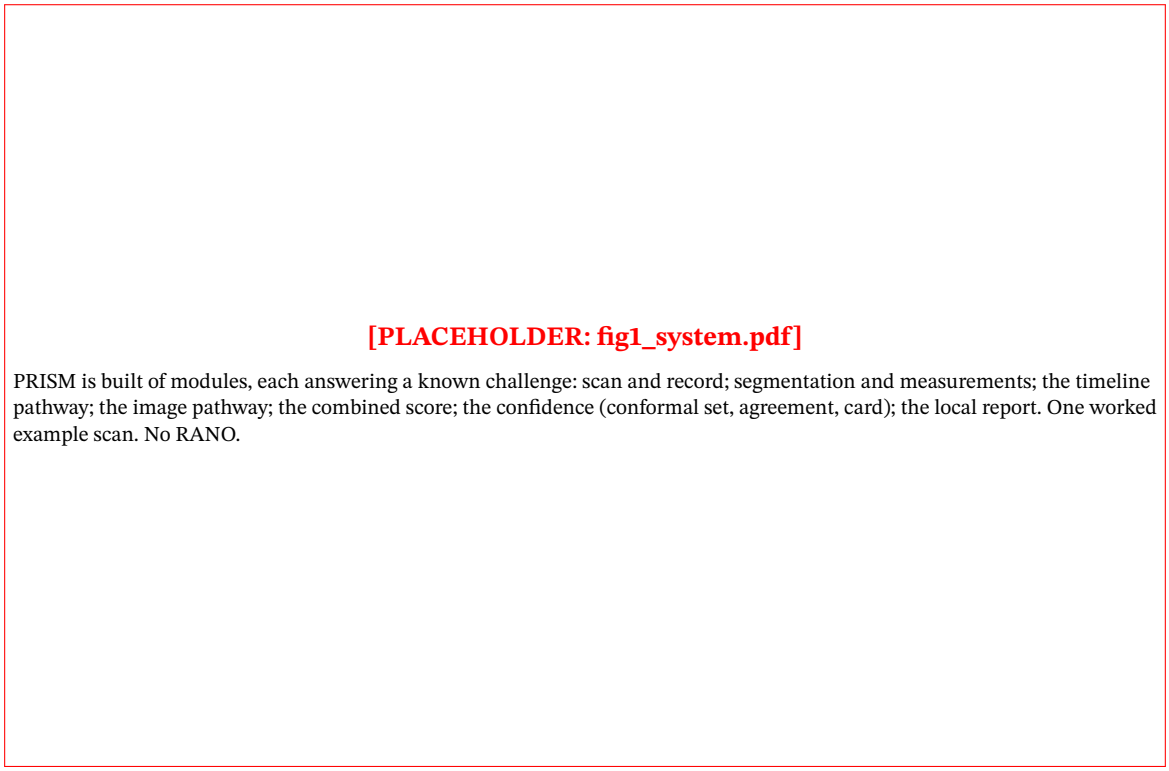

Figure 1. PRISM is built of modules, each answering a known challenge. [legend]

[PLACEHOLDER: fig2_transport.pdf]

AUC by pathway (clinical alone, timeline, images, combined) and cohort (MU internal; LUMIERE and UCSF locked; UCSF-trained as a reference).

Figure 2. The pathways hold at two external centres without refitting. [legend]

[PLACEHOLDER: fig3_evidence.pdf]

(A) MU: the timeline's gain lies between phases of care. (B) UCSF: enhancing volume alone matches the image model. (C) The image-model ladder (exploratory).

Figure 3. What each score rests on depends on the cohort. [legend]

[PLACEHOLDER: fig4_confidence.pdf]

(A) Conformal coverage at the 80% level by cohort and pathway. (B) AUC with and without the disagreement warning. (C) The reliability card: situations by cohort with their evidence tier.

Figure 4. The confidence keeps its coverage and points to doubtful scans. [legend]

[PLACEHOLDER: fig5_report.pdf]

One rendered report (segmentation figure, pathways, combined score, confidence) and faithfulness by cohort.

Figure 5. The report states the evidence, faithful to the record. [legend]